



Software Design and Analysis

Final Project Report

ARTEMIS

Advanced Real-time Tracking & Emergency Mission Integrated System

Submitted by:

Malhan Bin Faisal - 24i0651
Rehan Ayub - 24i0631
Moiz Khalid - 24i0674

Submitted to:
Mrs Afia Zafar

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1. Introduction

1.1 Purpose

This document specifies the software requirements and architectural design for the ARTEMIS Command Center, version 1.0. This Software Requirements Specification (SRS) covers the complete scope of the client-side desktop application, encompassing the user interface, database integration, threat assessment logic, and deployment architecture.

1.2 Product Scope

The ARTEMIS Command Center is a comprehensive, JavaFX-based desktop software solution designed for Security Operations Centers (SOCs). Its primary purpose is to facilitate real-time global emergency response and intelligence gathering. The software replaces fragmented communication tools by centralizing target dossiers (Digital Shadow), dynamic mission tracking (Live Operations Radar), and tactical asset deployment (Dispatch Center) into a single, highly secure, high-trust user interface.

1.3 Title

ARTEMIS Command Center: Global Emergency Response & Intelligence System. The project aims to provide an immediate, structured digital command environment to monitor, assess, and resolve high-stakes personnel emergencies globally.

1.4 Objectives

- To deliver a centralized, secure authentication and intelligence retrieval gateway.
- To automate the classification of incoming distress signals using an integrated AI Threat Assessment dictionary.
- To visualize active operations through a dynamic tactical radar interface.
- To streamline the deployment of proximity assets to active crisis zones.

1.5 Problem Statement

Modern emergency response coordination frequently suffers from "Component Bleed" and fragmented data silos, where personnel itineraries, live tracking, and dispatch communications are handled across disjointed systems. This manual fragmentation delays critical response times and increases cognitive load on operators. ARTEMIS mitigates this by providing a unified, data-driven command interface that automatically aggregates intelligence and prioritizes active missions based on algorithmic threat assessment.

2. Overall Description

2.1 Product Perspective

ARTEMIS is a new, self-contained software product. It operates as a localized client application communicating with an internal SQLite relational database to ensure high-speed data retrieval and absolute data privacy. It is designed with a "Ghost Protocol" enterprise aesthetic to reduce operator eye strain during extended monitoring sessions.

2.2 Product Functions

- **Identity Verification:** Secure user authentication and registration.
- **Tactical Radar:** Visual geographic representation of active responders and targets.
- **Dispatch Center:** Aggregation of active missions and available tactical assets for deployment.
- **Active SOS Feed:** Real-time logging and AI-driven classification of incoming distress signals.
- **Digital Shadow:** Deep-dive intelligence retrieval for specific member IDs.

2.3 List of Use Cases

1. Register Travel Itinerary
2. Trigger Emergency SOS
3. Retrieve Member Digital Shadow
4. Verify Emergency Status
5. Classify Mission Type
6. Query Proximity Assets
7. Authorize Extraction Mission
8. Track Live Exfiltration
9. Log Field Intelligence Update
10. Update Mission Lifecycle State
11. Generate Mission Audit Ledger
12. Manage System Global Configurations

2.4 Extended Use Cases: Fully Dressed Specifications

Use Case Section	Comment										
Use Case Name:	Register Travel Itinerary										
Scope:	Project ARTEMIS										
Level:	User-Goal										
Primary Actor:	Member										
Stakeholders and Interests:	Member, System Admin										
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. Member enters travel dates and hotel details.</td><td></td></tr> <tr> <td></td><td>2. System validates data and stores in SQL/Oracle database.</td></tr> <tr> <td>3. Member identifies local partners.</td><td></td></tr> <tr> <td></td><td>4. System links data to Member profile for SOS retrieval.</td></tr> </table>	Actor Action	System Response	1. Member enters travel dates and hotel details.			2. System validates data and stores in SQL/Oracle database.	3. Member identifies local partners.			4. System links data to Member profile for SOS retrieval.
Actor Action	System Response										
1. Member enters travel dates and hotel details.											
	2. System validates data and stores in SQL/Oracle database.										
3. Member identifies local partners.											
	4. System links data to Member profile for SOS retrieval.										
Extensions	1. Invalid date format. 2. Database connection failure.										
Pre-conditions	Member is logged into the system.										
Post conditions	Itinerary is successfully linked to Member ID.										

Use Case Section	Comment								
Use Case Name:	Trigger Emergency SOS								
Scope:	Project ARTEMIS								
Level:	User-Goal								
Primary Actor:	Member								
Stakeholders and Interests:	Member, Coordinator, Responders.								
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. Member presses the "Panic Button" on mobile app.</td><td></td></tr> <tr> <td></td><td>2. System captures GPS, altitude, and timestamp.</td></tr> <tr> <td></td><td>3. System alerts Command Center and creates Mission ID.</td></tr> </table>	Actor Action	System Response	1. Member presses the "Panic Button" on mobile app.			2. System captures GPS, altitude, and timestamp.		3. System alerts Command Center and creates Mission ID.
Actor Action	System Response								
1. Member presses the "Panic Button" on mobile app.									
	2. System captures GPS, altitude, and timestamp.								
	3. System alerts Command Center and creates Mission ID.								
Extensions	1. GPS signal lost. 2. Network timeout.								
Pre-conditions	Member app has active location permissions.								
Post conditions	Mission ID generated and SOS alert is active.								

Use Case Section	Comment
Use Case Name:	Retrieve Member Digital Shadow
Scope:	Project ARTEMIS
Level:	Sub-function
Primary Actor:	System
Stakeholders and Interests:	Coordinator (Needs immediate data).

Main Success Scenario	Actor Action	System Response
		1. System identifies Member ID from SOS packet.
		2. System queries database for itineraries and contacts.
		3. System pushes data to Coordinator dashboard.
Extensions	1. No active itinerary found.	
Pre-conditions	Incoming SOS signal detected.	
Post conditions	Coordinator dashboard populated with member data.	

Use Case Section	Comment	
Use Case Name:	Verify Emergency Status	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Coordinator	
Stakeholders and Interests:	Coordinator, Member, Family/Partner.	
Main Success Scenario	Actor Action	System Response
	1. Coordinator selects alert from dashboard.	
		2. System shows contacts and hotel info.
	3. Coordinator calls Member/Contact.	
	4. Contact confirms the emergency.	
		5. System opens log dialogue for mission notes.
	6. Coordinator sets status to "Verified".	
		7. System enables Classification module.
Extensions	1. Unresponsive Member. 2. False alarm cancellation.	
Pre-conditions	SOS received; Digital Shadow retrieved.	
Post conditions	Mission marked as Verified or False Alarm.	

Use Case Section	Comment	
Use Case Name:	Classify Mission Type	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Coordinator	
Stakeholders and Interests:	Coordinator, Responders.	

Main Success Scenario	Actor Action	System Response
	1. Coordinator reviews incident details.	
	2. Coordinator selects classification (Medical/Security).	
		3. System filters Asset Registry for responders.
Extensions	1. Mixed crisis (Multiple tags applied).	
Pre-conditions	SOS is verified as authentic.	
Post conditions	Mission is classified and asset list is filtered.	

Use Case Section	Comment	
Use Case Name:	Query Proximity Assets	
Scope:	Project ARTEMIS	
Level:	Sub-function	
Primary Actor:	System	
Stakeholders and Interests:	Coordinator, Google Maps API.	
Main Success Scenario	Actor Action	System Response
		1. System pulls GPS of all Responders.
		2. System sends request to Maps Distance API.
		3. System calculates ETA for each asset.
		4. System ranks closest responders on dashboard.
Extensions	1. API Timeout (Revert to Haversine calculation).	
Pre-conditions	Mission classified; Google Maps API online.	
Post conditions	Proximity list displayed to Coordinator.	

Use Case Section	Comment	
Use Case Name:	Authorize Extraction Mission	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Coordinator	
Stakeholders and Interests:	Coordinator, Responder.	

Main Success Scenario	Actor Action	System Response
	1. Coordinator selects a specific Responder.	
	2. Coordinator reviews exfiltration route.	
	3. Coordinator clicks "Authorize Deployment".	
		4. System sends push notification to Responder.
		5. System updates state to "Dispatched".
Extensions	1. Responder rejects mission.	
Pre-conditions	Assets identified and ranked.	
Post conditions	State changed to Dispatched; Responder notified.	

Use Case Section	Comment	
Use Case Name:	Track Live Exfiltration	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Coordinator	
Stakeholders and Interests:	Coordinator, Responder, Google Maps API.	
Main Success Scenario	Actor Action	System Response
	1. Coordinator opens "Live GRID" map.	
		2. System initiates telemetry polling.
		3. System calls Maps API to plot markers.
	4. Coordinator monitors distance and ETA.	
		5. System logs path every 10 seconds.
Extensions	1. Signal Loss. 2. Rerouting required.	
Pre-conditions	Mission state is "Dispatched" or "In-Progress".	
Post conditions	Mission path is fully logged and markers are active.	

Use Case Section	Comment	
Use Case Name:	Log Field Intelligence Update	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Responder	
Stakeholders and Interests:	Responder, Coordinator.	

Main Success Scenario	Actor Action	System Response
	1. Responder inputs text update.	
	2. Responder attaches situational media.	
		3. System updates Coordinator war-room feed.
		4. System saves entry into SQL history table.
Extensions	1. Upload failure due to low network signal.	
Pre-conditions	Responder is active on a mission.	
Post conditions	Intelligence data is persistently stored.	

Use Case Section	Comment	
Use Case Name:	Update Mission Lifecycle State	
Scope:	Project ARTEMIS	
Level:	Sub-function	
Primary Actor:	Coordinator	
Stakeholders and Interests:	Coordinator, Member, Family.	
Main Success Scenario	Actor Action	System Response
	1. Coordinator receives milestone update.	
	2. Coordinator updates state to "Secured".	
		3. System triggers notifications to family/club.
		4. System locks mission once state is "Closed".
Extensions	1. Unauthorized state transition attempt.	
Pre-conditions	Active mission is in progress.	
Post conditions	State machine is updated and notifications sent.	

Use Case Section	Comment	
Use Case Name:	Generate Mission Audit Ledger	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	System	
Stakeholders and Interests:	System Admin, Institutional Review.	

Main Success Scenario	Actor Action	System Response
		1. System aggregates GPS pings and logs.
		2. System compiles report of authorization steps.
		3. System generates PDF/Digital Ledger.
		4. System flags deviations from EAP.
Extensions	1. Incomplete log data.	
Pre-conditions	Mission is in "Closed" state.	
Post conditions	Audit ledger is available for download/export.	

Use Case Section	Comment	
Use Case Name:	Manage System Global Configurations	
Scope:	Project ARTEMIS	
Level:	User-Goal	
Primary Actor:	Admin	
Stakeholders and Interests:	Admin	
Main Success Scenario	Actor Action	System Response
	1. Admin logs into config portal.	
	2. Admin updates "Threat Level" for country.	
	3. Admin adds new Responder Asset type.	
		4. System updates global risk database.
Extensions	1. Database write error.	
Pre-conditions	Admin has active credentials.	
Post conditions	Global settings updated across system.	

2.5 Use Case Diagram



3. Other Nonfunctional Requirements

3.1 Performance Requirements

- **Response Time:** Database queries for the Digital Shadow and Dispatch Center must populate the user interface in under 1.5 seconds to ensure operational fluidity.
- **UI Rendering:** The JavaFX application must maintain smooth rendering of the Live Operations Radar without thread-blocking during data retrieval.

3.2 Safety Requirements

- **Data Persistence:** In the event of a system crash or power failure, the embedded SQLite database must guarantee that no active mission data or itinerary updates are lost or corrupted.
- **Validation Hardening:** The system must strictly enforce mandatory data fields (e.g., Safe-house Address) to ensure extraction teams are never deployed blindly.

3.3 Security Requirements

- **Authentication:** The system strictly requires valid credential verification before granting access to the overarching Command Center.
- **Data Isolation:** Operational data and personnel intelligence are strictly compartmentalized within the application's runtime environment.

3.4 Software Quality Attributes

- **Usability:** The interface adheres to "High-Trust Enterprise UI" standards (Ghost Protocol theme). It utilizes a top-bar/left-bar hybrid navigation rail to separate global controls from tactical tools, minimizing cognitive overhead.
- **Maintainability:** The codebase adheres to strict CSS class styling (`aegis-theme.css`) to prevent Modena bleed and ensure rapid future visual iterations without altering core Java logic.

3.5 Business Rules

- The system must never overwrite an existing user's ID during itinerary updates; it must utilize precise SQL `UPDATE` operations tied to the current active session state.
- A mission cannot be marked as "Completed" until an asset has been successfully dispatched.

3.6 Operating Environment

The software is designed to run on any standard desktop operating system (Windows, macOS, Linux) equipped with the Java Runtime Environment (JRE) supporting JavaFX 17 or higher.

3.7 User Interfaces

- **Login Pane:** A high-contrast split-pane layout featuring an indigo-to-cyan gradient branding panel and a secure authentication zone.
- **Dashboard:** A dark-mode (`#111827` root base) tactical layout. Features transparent sidebar buttons, custom-styled combo boxes, and color-coded statistical labels.
- **Radar Pane:** A centralized tactical map utilizing a high-contrast neon green sonar aesthetic against a dark backdrop.

4. Domain Model

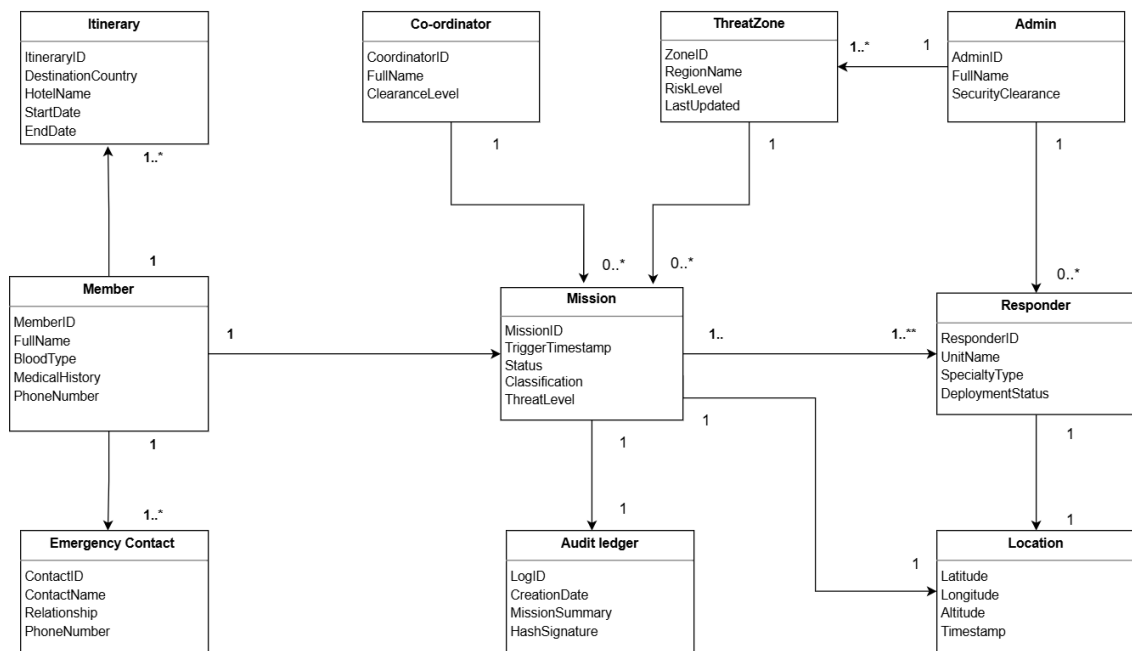


Figure 1: Domain Model Diagram

5. System Sequence Diagrams

5.1 Phase 1: Pre-Trip & Setup

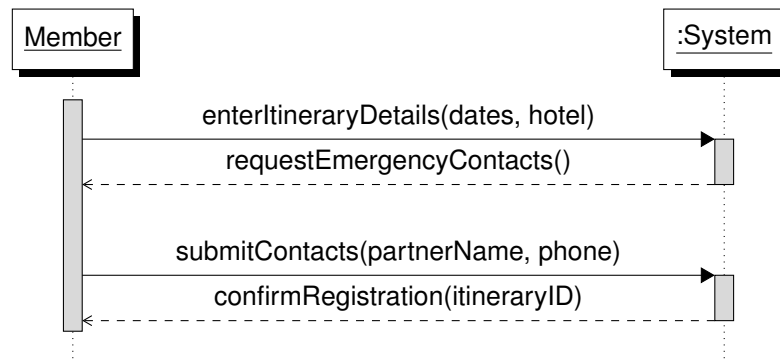


Figure 2: SSD: Register Travel Itinerary

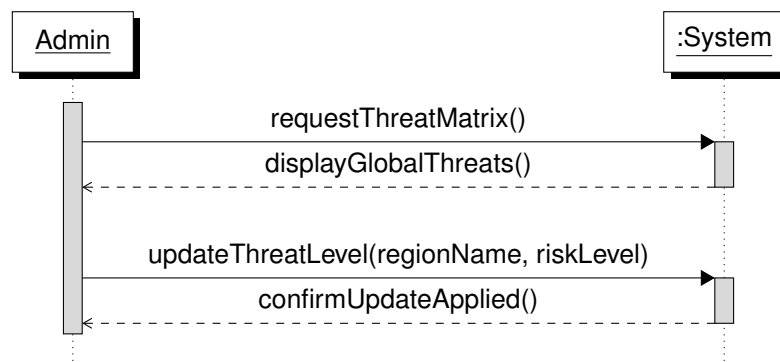


Figure 3: SSD: Manage System Global Configurations

5.2 Phase 2: The Emergency Trigger

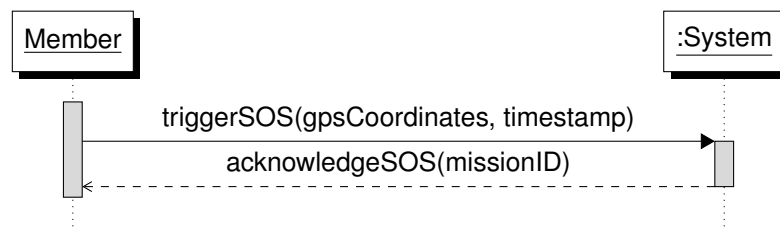


Figure 4: SSD: Trigger Emergency SOS

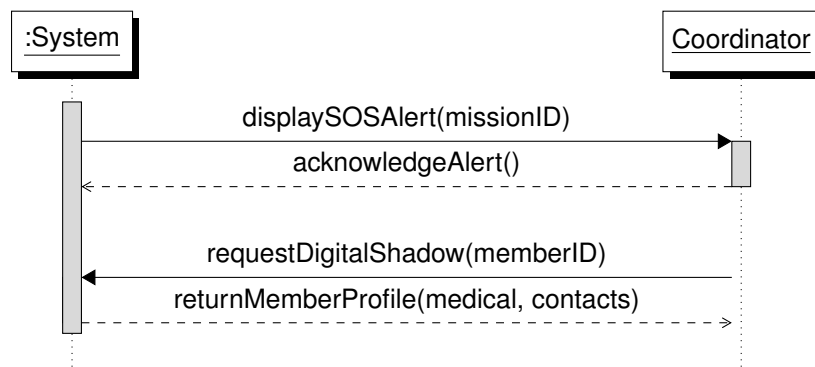


Figure 5: SSD: Retrieve Member Digital Shadow

5.3 Phase 3: The Verification Gate

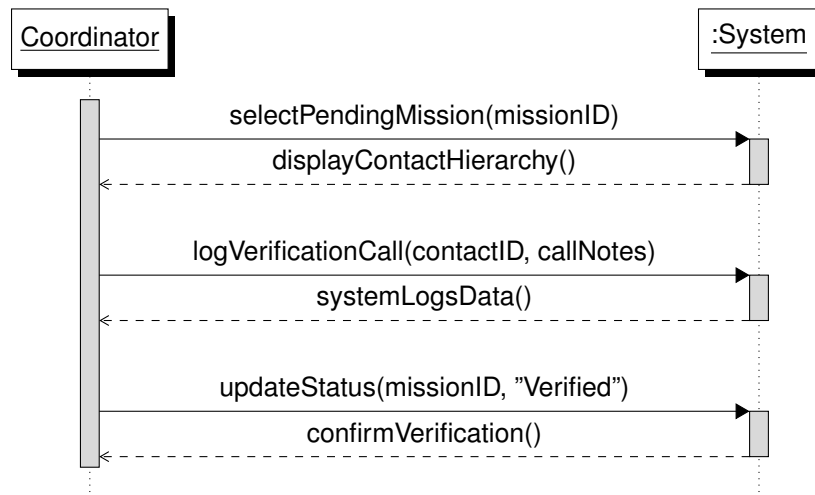


Figure 6: SSD: Verify Emergency Status

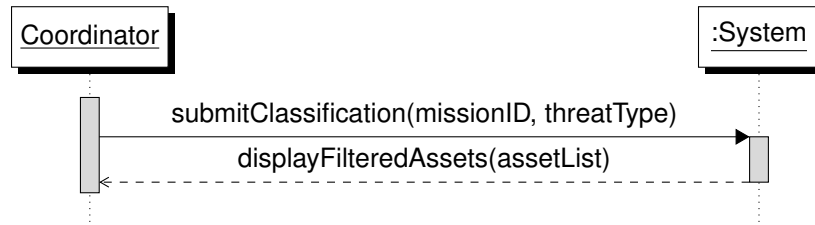


Figure 7: SSD: Classify Mission Type

5.4 Phase 4: Dispatch & Routing

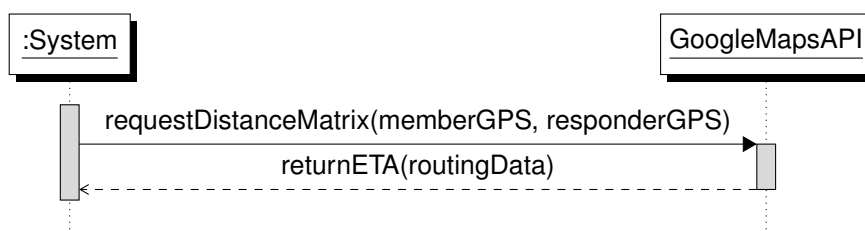


Figure 8: SSD: Query Proximity Assets

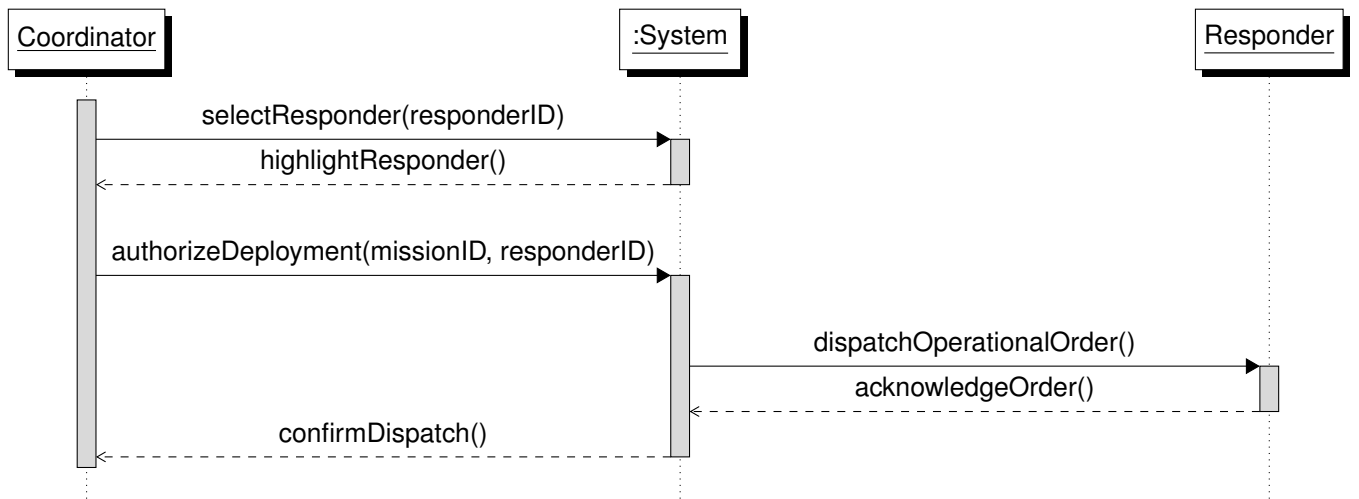


Figure 9: SSD: Authorize Extraction Mission

5.5 Phase 5: Active Operations

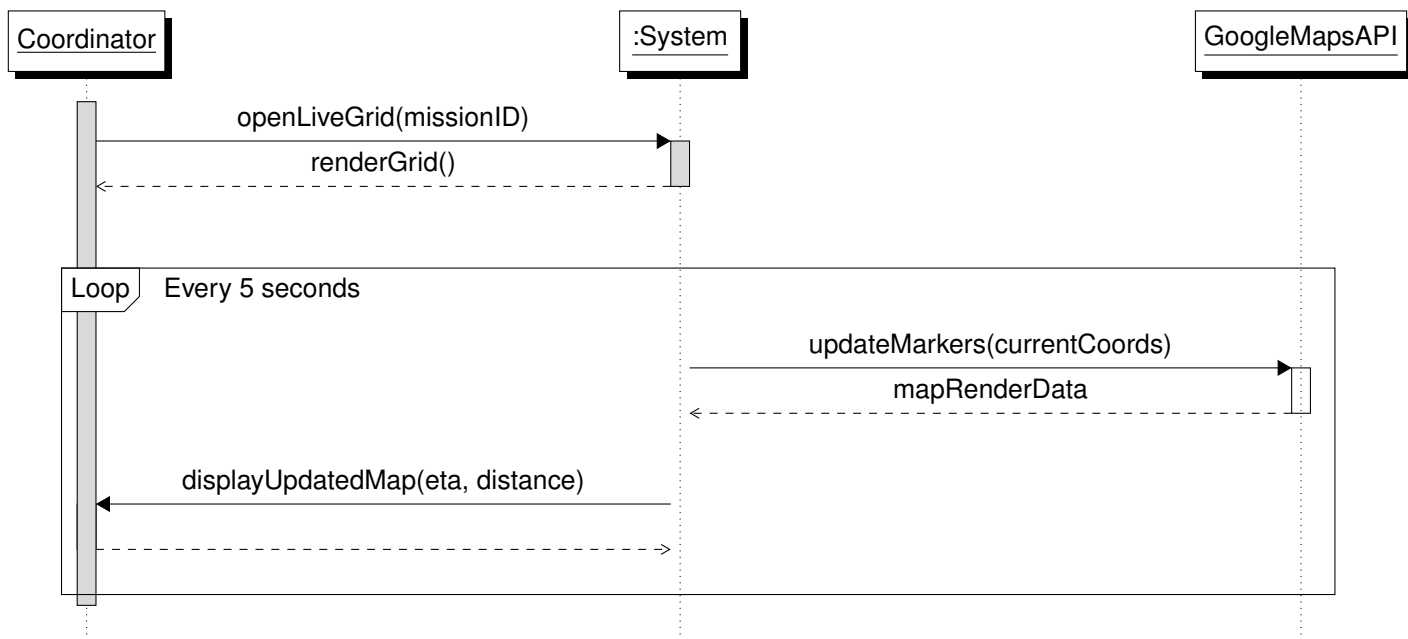


Figure 10: SSD: Track Live Exfiltration

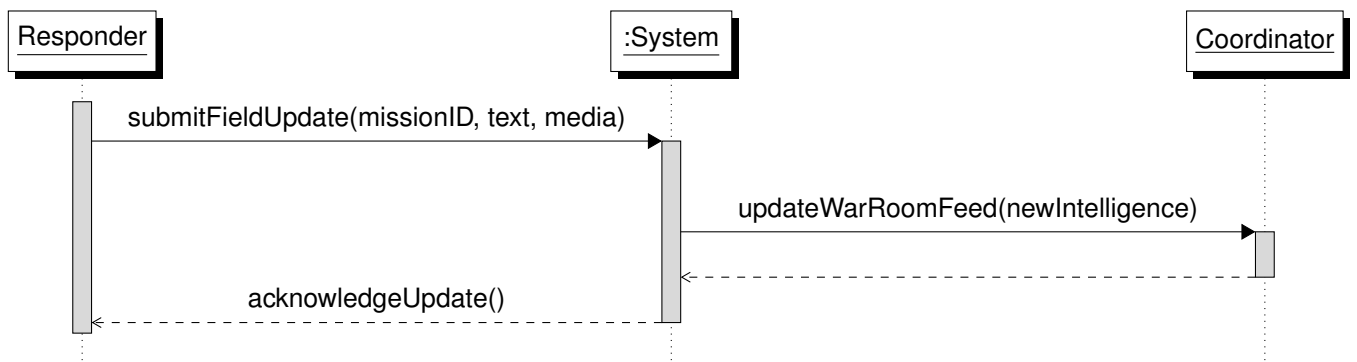


Figure 11: SSD: Log Field Intelligence Update

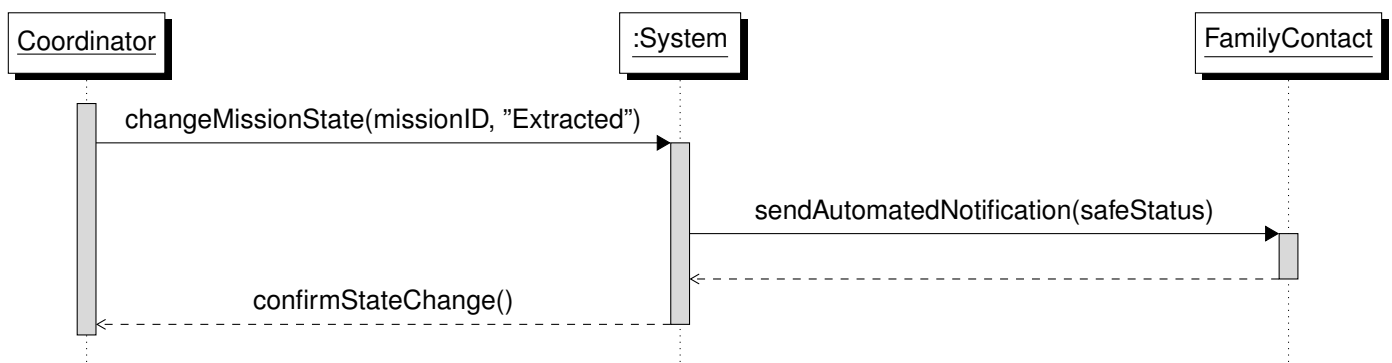


Figure 12: SSD: Update Mission Lifecycle State

5.6 Phase 6: Post-Mission

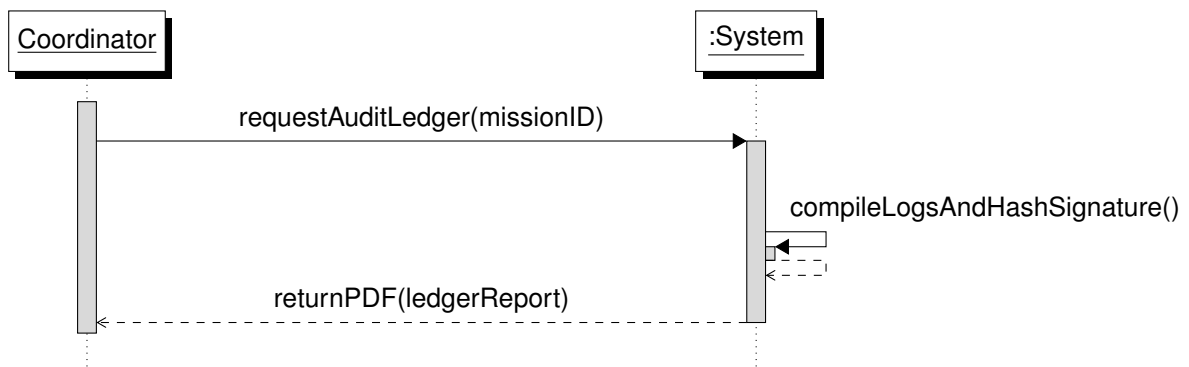


Figure 13: SSD: Generate Mission Audit Ledger

6. Sequence Diagram

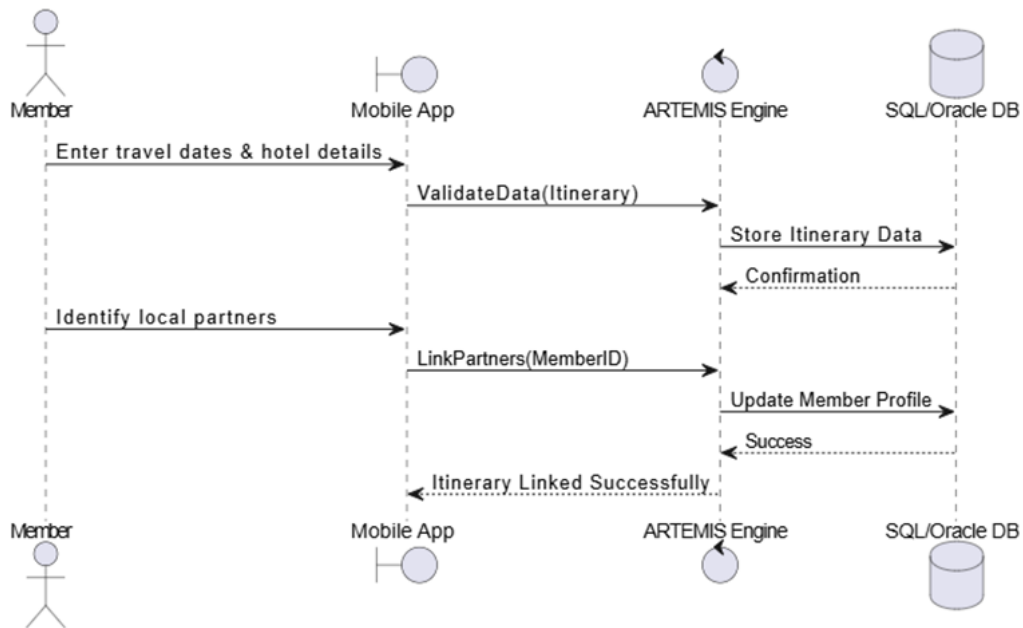


Figure 14: Sequence Diagram: Register Travel Itinerary

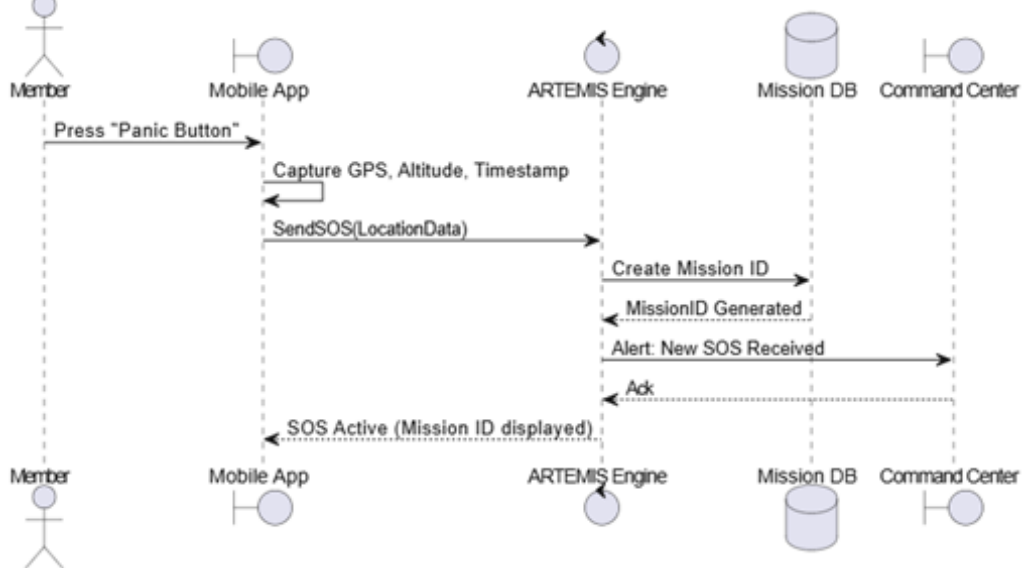


Figure 15: Sequence Diagram: Trigger Emergency SOS

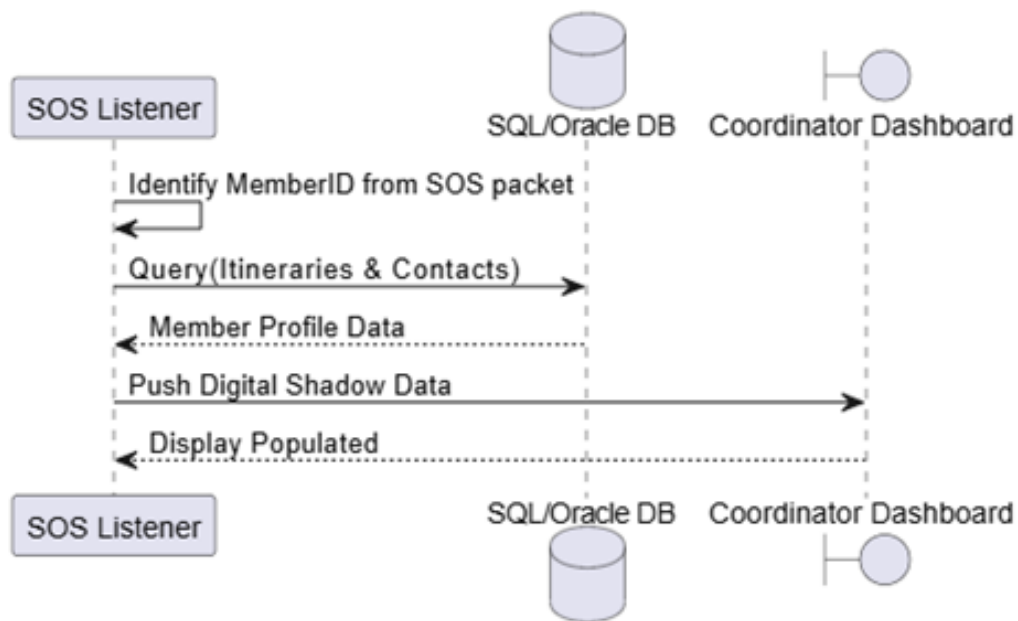


Figure 16: Sequence Diagram: Retrieve Member Digital Shadow

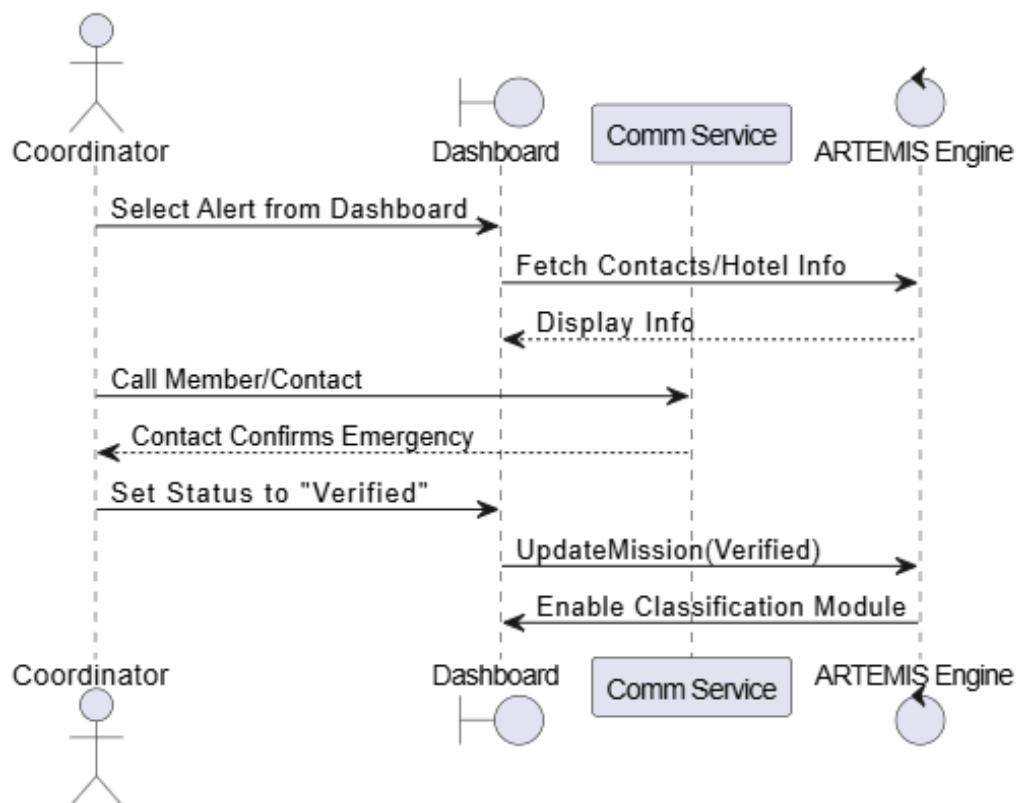


Figure 17: Sequence Diagram: Verify Emergency Status

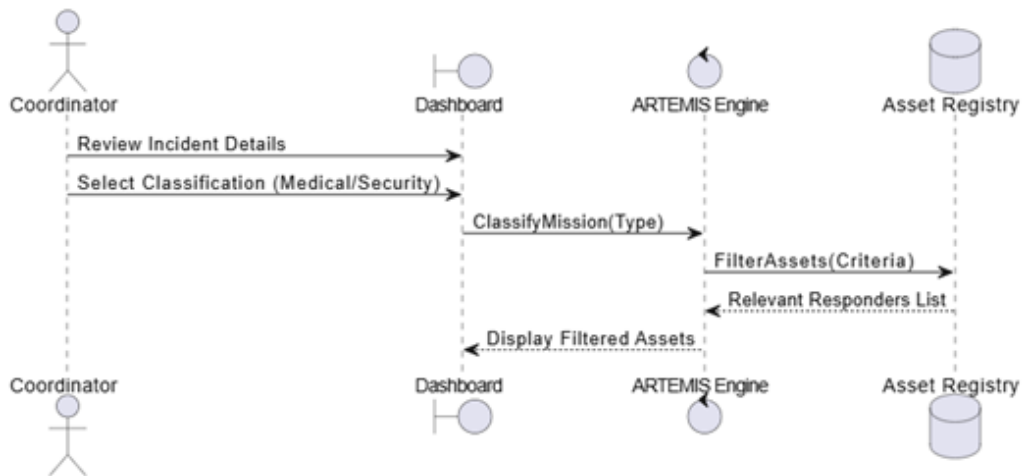


Figure 18: Sequence Diagram: Classify Mission Type

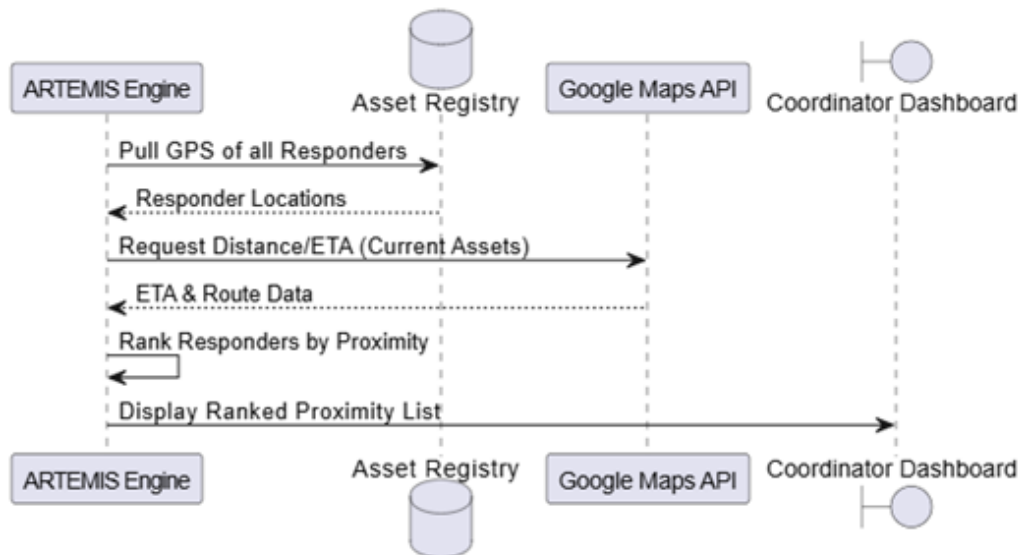


Figure 19: Sequence Diagram: Query Proximity Assets

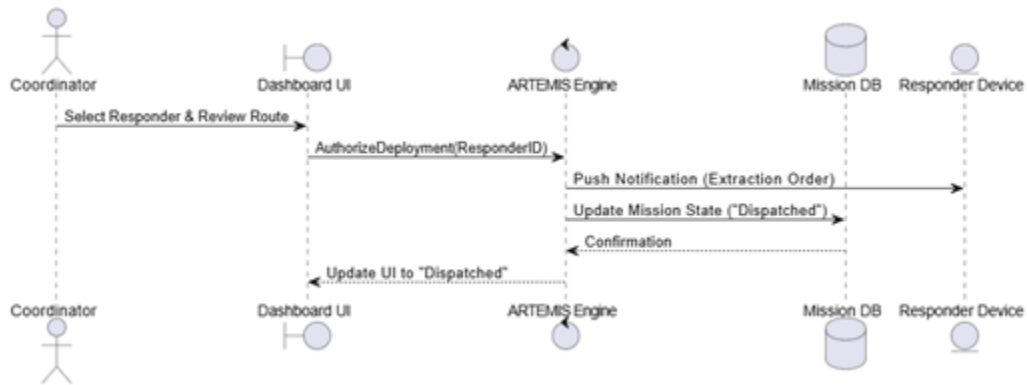


Figure 20: Sequence Diagram: Authorize Extraction Mission

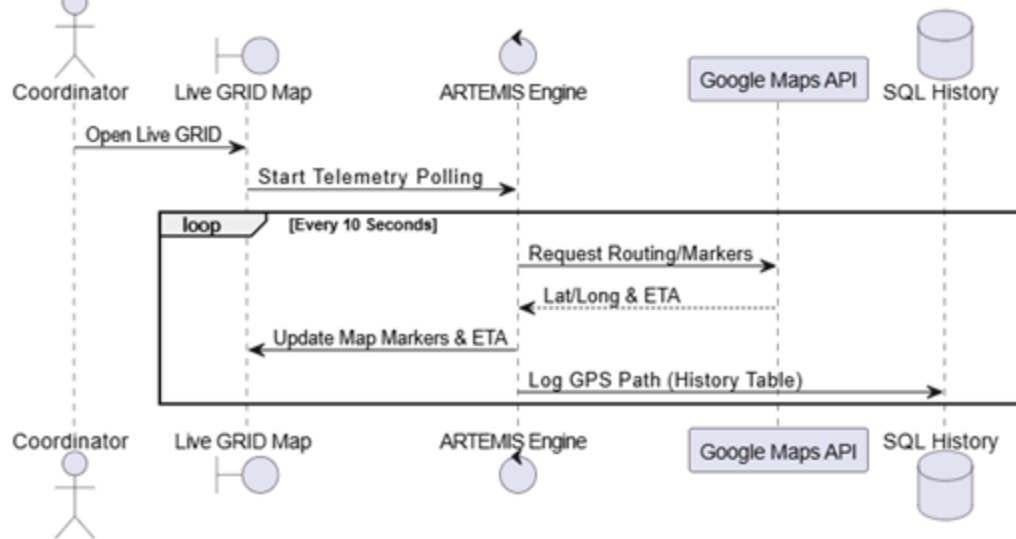


Figure 21: Sequence Diagram: Track Live Exfiltration

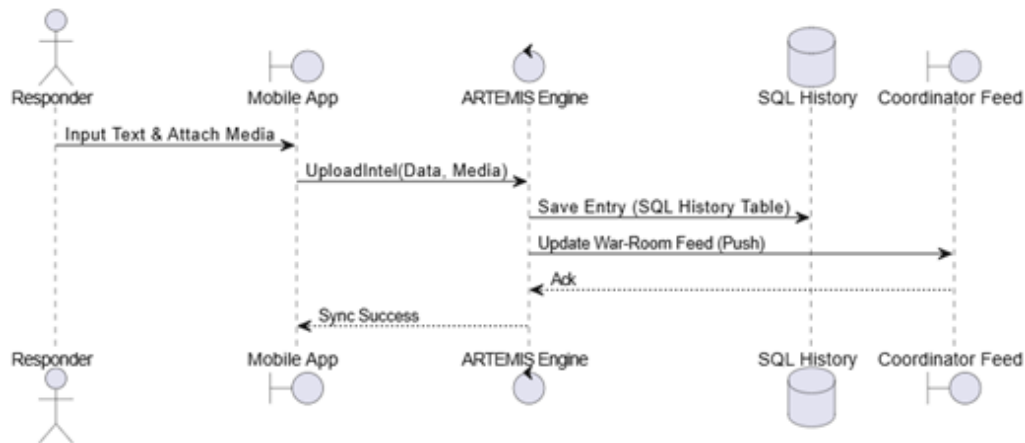


Figure 22: Sequence Diagram: Log Field Intelligence Update

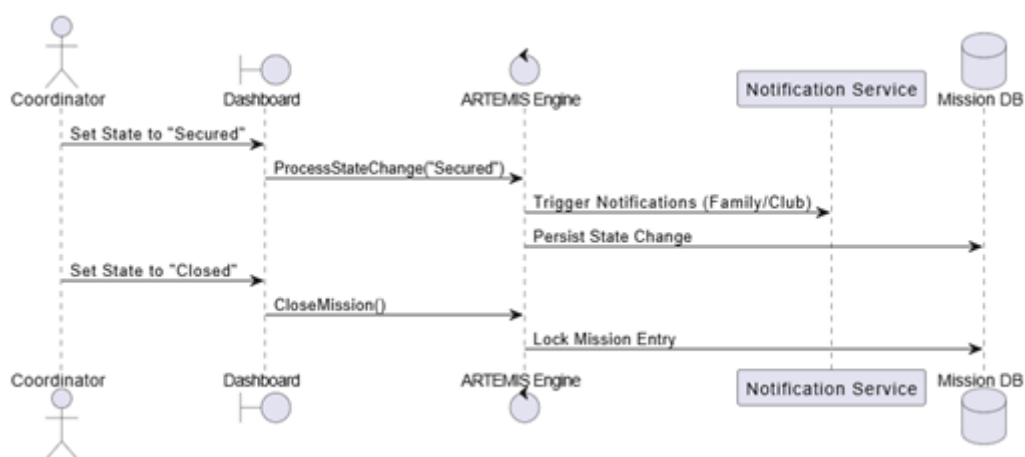


Figure 23: Sequence Diagram: Update Mission Lifecycle State

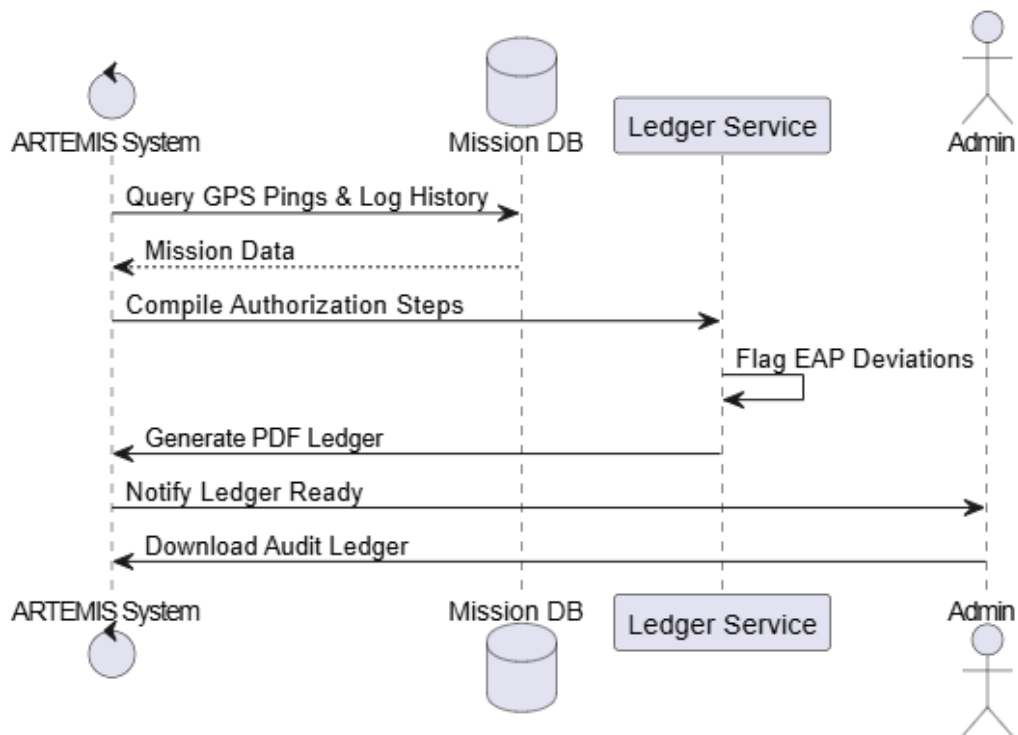


Figure 24: Sequence Diagram: Generate Mission Audit Ledger

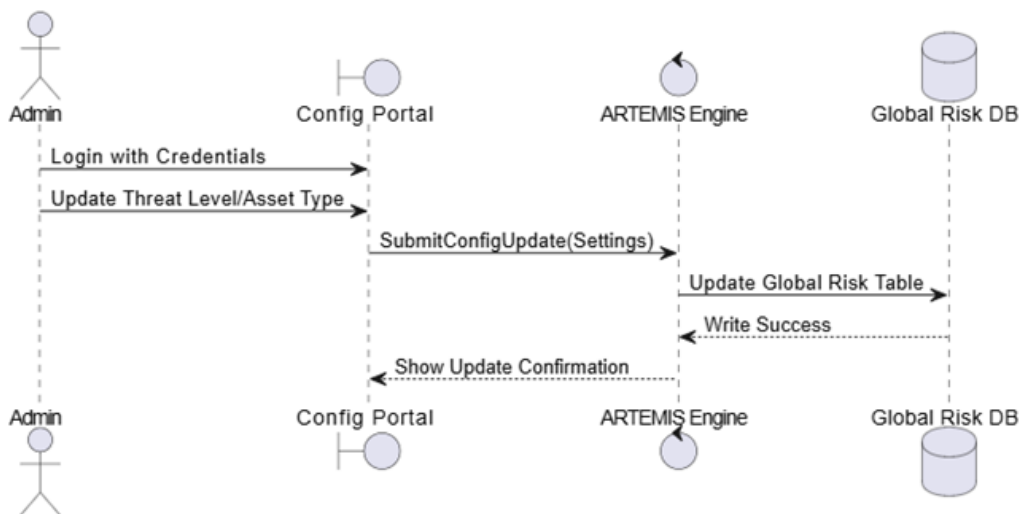


Figure 25: Sequence Diagram: Manage System Global Configurations

7. Class Diagram

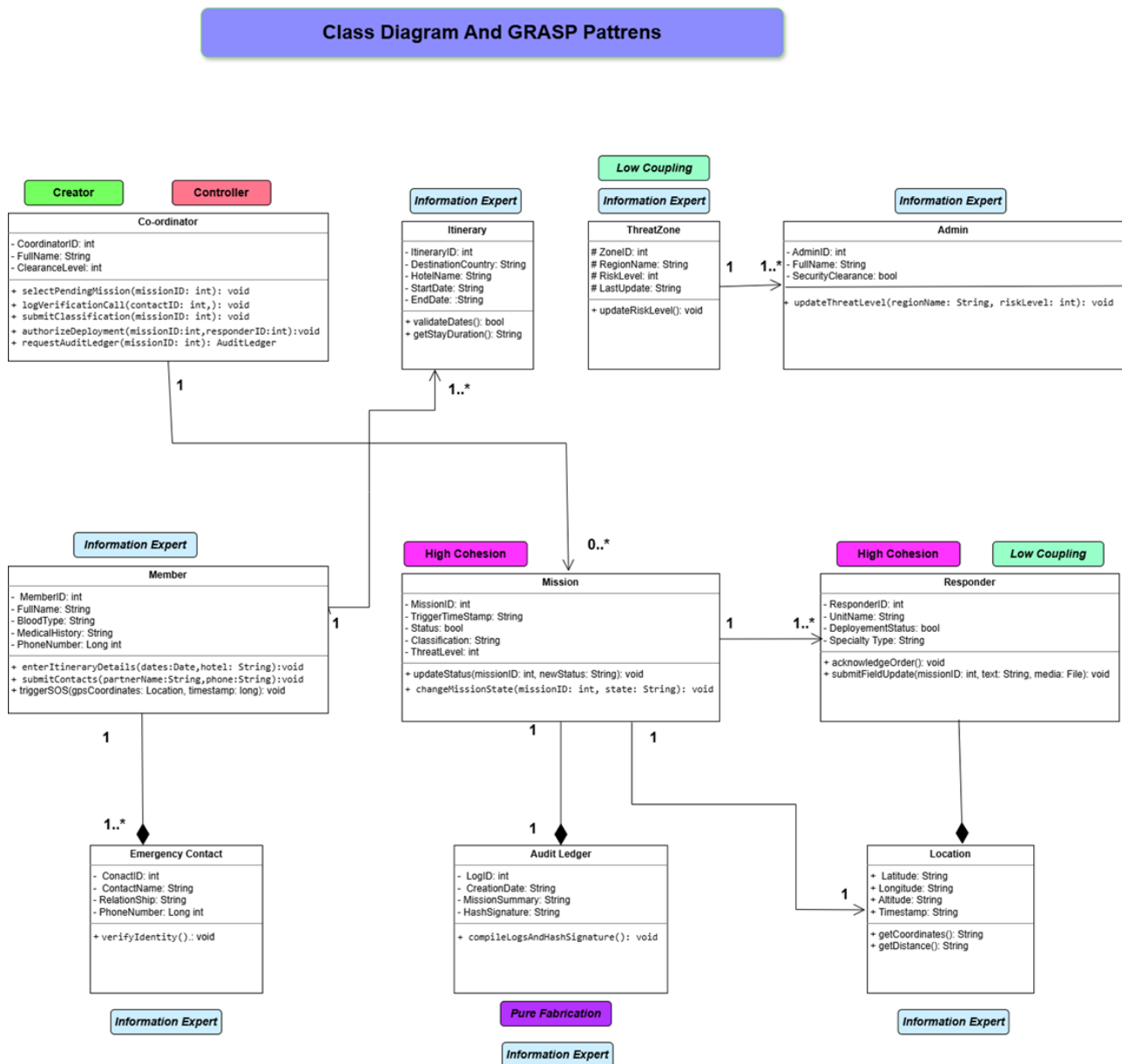


Figure 26: Class Diagram